# Activity 4: Select Controls for the Aster Robotics Handling Cell

**Objective:** Match grounding, ionisation, shielding, handling and monitoring controls to both human and automated exposure pathways.

## Scenario

A robot transfers packaged ICs from plastic tubes to a grounded nest. Operators also perform manual rework. The team proposes more wrist straps, but field readings near the tube exit remain high.

## Roles

- automation engineer
- ESD specialist
- maintenance supervisor

## Detailed procedure

1. Map every charged-person, charged-device and charged-machine pathway through the cell.
2. Identify where grounding is effective and where it is ineffective because the material is insulating.
3. Select personnel grounding for normal operation, setup and manual rework.
4. Select bonding and ground monitoring for robot fixtures, nests, tools and carts.
5. Evaluate whether the plastic tubes are essential and document the decision.
6. Define ioniser type, placement, airflow constraints and the required work-position checks.
7. Choose packaging/transfer methods that preserve shielding until entry into the protected process.
8. Define alarms, interlocks or stop conditions for loss of ground or ionisation.
9. Create a preventive-maintenance plan with cleaning and verification intervals.
10. Prepare a control-selection matrix linking each risk to control, owner, evidence and residual risk.
11. Walk through a changeover and maintenance intervention to find temporary exposures.
12. Present the selected design and respond to a cost-versus-risk challenge from management.

## Questions

- Why will adding wrist straps not solve the tube-exit field?
- Which controls address CDM risk?
- What should happen when an ioniser check fails?

## Evidence to submit

- Exposure-path map
- Control-selection matrix
- Monitoring/stop-condition plan
- Maintenance plan

## Acceptance criteria

- [ ] Controls cover human and automated pathways
- [ ] Insulators are addressed by an appropriate method
- [ ] Monitoring has explicit response actions
- [ ] Residual risk is stated

## Debrief

Explain which evidence changed your first assumption, which control is most reliable, and what follow-up measurement is needed.